Aligner IP Verification Project

RV Alignment Engine · APB Registers · Top-RTL · GDS Support

CPU-A	RV Alignment	CPU-B Interface	Aligner Core	Registers

AMT AXI4 MASTER	AXI4-MM PERIPHERAL	AXI-MM PERIPHERAL
512-bit data, 40-bit addr, 16-bit ID	32-bit data, 32-bit addr, 16-bit ID	32-bit data, 32-bit addr, 16-bit ID
INCR bursts, 4KB boundary	APB Agent, 4KB 32-bit region, 100 MHz	Shared with CPU-A Interface
Outstanding: 8 rd / 8 wr	Decode: 0x0000 Control, 0x0100 Status	Decode: 0x0200 Control, 0x0300 Status
1K range, single	Scratchpad: 256 Doublewords	Scratchpad: 256 Doublewords
1K range, single	Interrupt: Frame Done	Interrupt: Frame Done
Aligns 4MB frames	Reg Protection: Reg Locked	Protection: W-1-C

1. Project Overview

The project aligner and the ARMv8-A aligner IP block using SystemVerilog RTL and the Universal Verification Methodology (UVM) verification framework. The aligner IP performs real-time pixel alignment between two camera streams, applying per-channel gain, configurable alignment interval, and statistics frame as an AMT TX interface. The block is constructed as an APB register interface that exists on the CPU-A AXI4-MM interface, a CPU-B AXI4-MM interface, and a dedicated AXI4-MM master port for DMA-style frame output. The design is verified using a UVM testbench with a scoreboard, coverage collector, and SVA assertions, and the RTL is synthesized through Cadence Genus and Cadence Innovus.

1.1 Design Scope

- 32-bit pixel datapath (unsigned Q0.16 gain with offset and clamp to [0, 65535])
- Dual-camera alignment engine — handles split-line RV — multiplier TX and merge-buffer RV — one TX, scatterwrite
- Frame-interval gating: 1 frame in N (N programmable, 1 to 255) passes through
- Dual-port scoreboard RV8 and TX8 with optional gain-aware CRC support
- 8-slot outstanding AXI4 master (CPU-B) with AWID/ARID for out-of-order transaction monitoring
- Frame-done IRQ, sticky error flags, and a 256-word scratchpad
- Interrupt generation on FRAME_DONE and AXI4 bus error conditions

1.2 Technology Stack

Category	Tools
RTL	SystemVerilog IEEE 1800-2012
Verification Methodology	UVM 1.0 — uvm_agent, uvm_scoreboard, uvm_monitor, uvm_reg_adapter, uvm_reg_predictor, uvm_reg_block, uvm_reg_field, uvm_reg_map, uvm_sequence, uvm_sequence_item, uvm_driver, uvm_monitor, uvm_scoreboard, uvm_subscriber, uvm_coverpoint, uvm_cross, uvm_bin, uvm_ignore_bins, uvm_illegal_bins, uvm_wildcard_bins, uvm_cross_bins, uvm_cross_ignore_bins, uvm_cross_illegal_bins, uvm_cross_wildcard_bins, uvm_cross_binsof, uvm_cross_intersect, uvm_cross_with, uvm_cross_cross, uvm_cross_cross_bins, uvm_cross_cross_ignore_bins, uvm_cross_cross_illegal_bins, uvm_cross_cross_wildcard_bins, uvm_cross_cross_binsof, uvm_cross_cross_intersect, uvm_cross_cross_with, uvm_cross_cross_cross, uvm_cross_cross_cross_bins, uvm_cross_cross_cross_ignore_bins, uvm_cross_cross_cross_illegal_bins, uvm_cross_cross_cross_wildcard_bins, uvm_cross_cross_cross_binsof, uvm_cross_cross_cross_intersect, uvm_cross_cross_cross_with, uvm_cross_cross_cross_cross, uvm_cross_cross_cross_cross_bins, 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